

GANGASAGAR HL

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EDUCATION

M.Tech in Computer Science, International Institute of Information Technology Bangalore July 2024 – Present
GPA: 3.58/4

Bachelor of Engineering in Computer Science, Sapthagiri College Engineering August 2017 – July 2021
GPA: 3.59/4

EXPERIENCE

IoT Intern, SmartBridge, Remote [Link](#) Aug 2020

- Built end-to-end smart surveillance system using IBM Watson IoT for real-time monitoring and control.
- Integrated image capture with Cloud Object Storage, Cloudant DB, and Watson Visual Recognition for threat detection.
- Orchestrated flows via Node-RED and triggered Fast2SMS mobile alerts for unknown face detection.

PROJECTS

Krishisahayogi — OpenCV, Flask, LSTM, ARIMA, Random Forest, KNN [Link](#) July 2021

- Developed a full-stack agri-assistant platform for crop selection, soil analysis, and market forecasting.
- Used OpenCV + Random Forest to predict soil pH from images; applied KNN and weather forecasts for crop recommendation.
- Forecasted crop prices using real APMC data with LSTM + ARIMA to help farmers maximize profitability.

Security System — Python, Flask, PyTorch, LangChain, TensorFlow Lite, FAISS, Raspberry Pi 3B+ [Link](#) Apr 2025

- Deployed MobileNetV3 (TFLite) on Raspberry Pi for real-time object detection and alert-based video streaming.
- Integrated BLIP, T5, LLaMA 3.2 to auto-summarize alerts with Twilio/WhatsApp/Gmail-based notifications.
- Built RAG pipeline using FAISS to index/summarize daily visual data and served via REST API.

Heart Disease Detection — Airflow, MLFlow, Sklearn, Docker, AWS (ECR, EC2, S3), K8s, Actions [Link](#) May 2025

- Orchestrated ETL with Airflow (GitHub → MongoDB Atlas); automated CI/CD via Actions, Docker, ECR, EC2.
- Stored best models on S3; enabled public EC2-based access for inference and online training.
- Simulated Kubernetes deployment locally for horizontally scalable ML workloads.

Transformers Arena — PyTorch, LoRA, Huggingface Hub, Qwen Transformer, Raspberry Pi 3B+ May 2025

- **GPT-2 Model:** Built custom GPT-2 in PyTorch for Raspberry Pi via hyperparameter tuning; published to PyPI as `sagar-neuron-gpt2`. Developed `sagar-neuron-gpt2`, a 38MB GPT-2 variant with perplexity 20 for edge inference; [Link](#).
- **One Word VQA:** Fine-tuned Qwen Vision Transformer using LoRA for one-word Visual QA on the Amazon-Berkeley dataset. [Link](#).

TeleManas Project (Ongoing) — Streamlit, LangChain, LangGraph, LoRA, MySQL, SQL [Link](#) Jan 2025 – Present

- Ongoing government-funded project under college for the national TeleManas mental health initiative.
- Building an intelligent dashboard that converts natural language queries into visual insights using LangChain.
- Automates data analysis to address policy-related queries from Government of India stakeholders.

PUBLICATIONS

Predicting Soil pH using Mobile Captured Picture [Link](#) March 2021

Gangasagar HL, Jovin Dsouza, Bhagyashree B Yargal, Arun Kumar SV, Anuradha Badage. *International Research Journal of Engineering and Technology (IRJET)*, Vol. 08, Issue 07, July 2021.

TECHNICAL SKILLS

Languages: Python, C, C++ **Technologies/Frameworks:** TensorFlow, PyTorch, Docker, Git, GitHub, GitHub Actions, Flask, MySQL, SQL, ML Flow, Apache Airflow, Raspberry Pi

RELEVANT COURSEWORK

Software Systems Concrete Mathematics Machine Learning Mathematics For ML Enterprise System Development
Visual Recognition Multi Objective Machine Learning Project Elective

LEADERSHIP / EXTRACURRICULAR

Volunteered in institute-level activities and events under the Aikyam student body.

Spring 2020 – Present

District Level Table Tennis Winner.